20261116 Quiz

for (MATH210) Applied Complex Variables

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Exercise 77.1. Find the residue at $z = 0$ of the function

(a) $\dfrac{1}{z+z^2}$;

(b) $z\cos\left(\dfrac{1}{z}\right)$;

(c) $\dfrac{z-\sin z}{z}$;

(d) $\dfrac{\cot z}{z^4}$;

(e) $\dfrac{\sinh z}{z^4(1-z^2)}$.

Solution. In each case, the residue at the origin is the coefficient of z^{-1} in the Laurent series.

(a) Using the geometric series,

$$\frac{1}{z+z^2} = \frac{1}{z}\frac{1}{1+z} = \frac{1}{z} - 1 + z - \cdots \qquad (0 < |z| < 1).$$

Hence $\boxed{\operatorname{Res}(f;0) = 1}$.

(b) The Laurent expansion is

$$\begin{aligned} z\cos\left(\frac{1}{z}\right) &= z\left(1 - \frac{1}{2!z^2} + \frac{1}{4!z^4} - \cdots\right) \\ &= z - \frac{1}{2z} + \frac{1}{4!z^3} - \cdots . \end{aligned}$$

Thus $\boxed{\operatorname{Res}(f;0) = -\dfrac{1}{2}}$.

(c) Since

$$\frac{z-\sin z}{z} = 1 - \frac{\sin z}{z} = \frac{z^2}{3!} - \frac{z^4}{5!} + \cdots ,$$

the origin is removable and $\boxed{\operatorname{Res}(f;0) = 0}$.

(d) From the Taylor series for sine and cosine,

$$\begin{aligned} \cot z &= \frac{1 - z^2/2! + z^4/4! - \cdots}{z(1 - z^2/3! + z^4/5! - \cdots)} \\ &= \frac{1}{z} - \frac{z}{3} - \frac{z^3}{45} + \cdots . \end{aligned}$$

Consequently,

$$\frac{\cot z}{z^4} = \frac{1}{z^5} - \frac{1}{3z^3} - \frac{1}{45z} + \cdots, \qquad \boxed{\operatorname{Res}(f;0) = -\frac{1}{45}}.$$

(e) Multiplying the series

$$\sinh z = z + \frac{z^3}{3!} + \cdots, \qquad \frac{1}{1-z^2} = 1 + z^2 + \cdots,$$

gives

$$\frac{\sinh z}{1-z^2} = z + \frac{7}{6}z^3 + \cdots.$$

Hence

$$\frac{\sinh z}{z^4(1-z^2)} = \frac{1}{z^3} + \frac{7}{6z} + \cdots, \qquad \boxed{\operatorname{Res}(f;0) = \frac{7}{6}}.$$

□

Exercise 77.2. Use Cauchy's residue theorem (Sec. 76) to evaluate the integral of each of these functions around the circle $|z| = 3$ in the positive sense:

(a) $\dfrac{\exp(-z)}{z^2}$;

(b) $\dfrac{\exp(-z)}{(z-1)^2}$;

(c) $z^2 \exp\left(\dfrac{1}{z}\right)$;

(d) $\dfrac{z+1}{z^2-2z}$.

Solution. Let C be the positively oriented circle $|z| = 3$. In each case, Cauchy's residue theorem gives

$$\int_C f(z)\,\mathrm{d}z = 2\pi i \sum_{a \text{ inside } C} \operatorname{Res}(f;a).$$

(a) At $z = 0$,

$$\frac{e^{-z}}{z^2} = \frac{1}{z^2} - \frac{1}{z} + \frac{1}{2} - \cdots.$$

The residue is -1, so $\boxed{\displaystyle\int_C \frac{e^{-z}}{z^2}\,\mathrm{d}z = -2\pi i}$.

(b) The only pole is the double pole $z = 1$. By the generalized Cauchy integral formula (Sec. 55),

$$\operatorname{Res}\left(\frac{e^{-z}}{(z-1)^2};1\right) = \frac{\mathrm{d}}{\mathrm{d}z}e^{-z}\bigg|_{z=1} = -\frac{1}{e}.$$

Thus $\boxed{\displaystyle\int_C \frac{e^{-z}}{(z-1)^2}\,\mathrm{d}z = -\frac{2\pi i}{e}}$.

(c) Expanding about the isolated singularity at $z = 0$,

$$z^2 e^{1/z} = z^2 \sum_{n=0}^{\infty} \frac{1}{n!z^n} = z^2 + z + \frac{1}{2} + \frac{1}{3!z} + \cdots.$$

The residue is $1/3! = 1/6$. Therefore

$$\boxed{\int_C z^2 e^{1/z}\,\mathrm{d}z = \frac{\pi i}{3}}.$$

(d) The two simple poles 0 and 2 lie inside C, with residues

$$\text{Res}(f;0) = \left.\frac{z+1}{z-2}\right|_{z=0} = -\frac{1}{2}, \qquad \text{Res}(f;2) = \left.\frac{z+1}{z}\right|_{z=2} = \frac{3}{2}.$$

Their sum is 1, so

$$\boxed{\int_C \frac{z+1}{z^2-2z}\,\mathrm{d}z = 2\pi i}.$$

□

Exercise 77.6. Suppose that a function f is analytic throughout the finite plane except for a finite number of singular points $z_1, z_2, \ldots, z_n$. Show that

$$\text{Res}(f;z_1) + \text{Res}(f;z_2) + \cdots + \text{Res}(f;z_n) + \text{Res}(f;\infty) = 0.$$

Solution. Choose a sufficiently large positively oriented circle C containing all the finite singular points. By Cauchy's residue theorem (Sec. 76),

$$\int_C f(z)\,\mathrm{d}z = 2\pi i \sum_{k=1}^{n} \text{Res}(f;z_k).$$

On the other hand, the definition of the residue at infinity and the theorem in Sec. 77 give

$$\int_C f(z)\,\mathrm{d}z = -2\pi i\,\text{Res}(f;\infty).$$

Equating these expressions and dividing by $2\pi i$ yields

$$\boxed{\sum_{k=1}^{n} \text{Res}(f;z_k) + \text{Res}(f;\infty) = 0.}$$

□

Exercise 77.7. Let the degrees of the polynomials

$$P(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n \qquad (a_n \neq 0)$$

and

$$Q(z) = b_0 + b_1 z + b_2 z^2 + \cdots + b_m z^m \qquad (b_m \neq 0)$$

be such that $m \geq n+2$. Use the theorem in Sec. 77 to show that if all of the zeros of $Q(z)$ are interior to a simple closed contour C, then

$$\int_C \frac{P(z)}{Q(z)}\,\mathrm{d}z = 0.$$

[Compare with Exercise 4(b).]

Solution. Set $f(z) = P(z)/Q(z)$; all its possible finite singularities are inside C. Writing $w = 1/z$, we obtain

$$\begin{aligned}\frac{1}{w^2} f\left(\frac{1}{w}\right) &= \frac{1}{w^2}\frac{P(1/w)}{Q(1/w)} \\ &= w^{m-n-2}\,\frac{a_n + a_{n-1}w + \cdots + a_0 w^n}{b_m + b_{m-1}w + \cdots + b_0 w^m}.\end{aligned}$$

Since $m - n - 2 \geq 0$ and the denominator has the nonzero value b_m at $w = 0$, this function is analytic at 0. In particular, its residue at 0 is zero. By the theorem in Sec. 77 (taking C positively oriented),

$$\boxed{\int_C \frac{P(z)}{Q(z)}\,\mathrm{d}z = 2\pi i \operatorname{Res}\left(\frac{1}{w^2} f\left(\frac{1}{w}\right); 0\right) = 0.}$$

The integral is also zero if C is taken with the opposite orientation.

□

Exercise 79.1. In each case, write the principal part of the function at its isolated singular point and determine whether that point is a removable singular point, an essential singular point, or a pole:

(a) $z \exp\left(\frac{1}{z}\right)$;

(b) $\frac{z^2}{1+z}$;

(c) $\frac{\sin z}{z}$;

(d) $\frac{\cos z}{z}$;

(e) $\frac{1}{(2-z)^3}$.

Solution. The principal part consists of the negative-power terms of the Laurent series at the indicated singular point (Sec. 78).

(a) At $z = 0$,

$$ze^{1/z} = z + 1 + \frac{1}{2!z} + \frac{1}{3!z^2} + \cdots.$$

The principal part is

$$\boxed{\sum_{n=2}^{\infty} \frac{1}{n!z^{n-1}}}.$$

It has infinitely many nonzero terms, so 0 is an *essential singular point.*

(b) At $z = -1$, put $w = z + 1$. Then

$$\frac{z^2}{1+z} = \frac{(w-1)^2}{w} = \frac{1}{w} - 2 + w.$$

The principal part is $\boxed{\frac{1}{z+1}}$, and -1 is a *simple pole.*

(c) At $z = 0$,

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots.$$

There are no negative powers. The principal part is $\boxed{0}$, and 0 is a *removable singular point.*

(d) At $z = 0$,

$$\frac{\cos z}{z} = \frac{1}{z} - \frac{z}{2!} + \frac{z^3}{4!} - \cdots.$$

The principal part is $\boxed{\frac{1}{z}}$, so 0 is a *simple pole.*

(e) At $z = 2$,

$$\frac{1}{(2-z)^3} = -\frac{1}{(z-2)^3}.$$

The principal part is $\boxed{-\dfrac{1}{(z-2)^3}}$, so 2 is a *pole of order* 3.

□

Exercise 79.3. Suppose that a function f is analytic at z_0, and write $g(z) = f(z)/(z - z_0)$. Show that

(a) if $f(z_0) \neq 0$, then z_0 is a simple pole of g, with residue $f(z_0)$;

(b) if $f(z_0) = 0$, then z_0 is a removable singular point of g.

Suggestion. As pointed out in Sec. 62, there is a Taylor series for $f(z)$ about z_0 since f is analytic there. Start each part of this exercise by writing out a few terms of that series.

Solution. Because f is analytic at z_0, its Taylor series has the form

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \cdots$$

near z_0. Dividing by $z - z_0$ gives, for $z \neq z_0$,

$$g(z) = \frac{f(z_0)}{z - z_0} + f'(z_0) + \frac{f''(z_0)}{2!}(z - z_0) + \cdots .$$

(a) If $f(z_0) \neq 0$, the principal part consists of the nonzero term $f(z_0)/(z - z_0)$. Hence z_0 is a simple pole and

$$\boxed{\text{Res}(g; z_0) = f(z_0).}$$

(b) If $f(z_0) = 0$, all negative-power terms vanish. The remaining Taylor series extends g analytically to z_0 by setting $g(z_0) = f'(z_0)$. Hence z_0 is a removable singular point.

□

Exercise 81.4. Find the value of the integral

$$\int_C \frac{3z^3 + 2}{(z-1)(z^2+9)}\, dz,$$

taken counterclockwise around the circle (a) $|z - 2| = 2$; (b) $|z| = 4$.

Solution. The integrand has simple poles at 1, $3i$, and $-3i$. The residue at 1 is

$$\text{Res}(f; 1) = \left.\frac{3z^3 + 2}{z^2 + 9}\right|_{z=1} = \frac{1}{2}.$$

For the other two poles,

$$\text{Res}(f; 3i) = \left.\frac{3z^3 + 2}{(z-1)(z+3i)}\right|_{z=3i} = \frac{5}{4} + \frac{49}{12}i,$$

and, since f has real coefficients,

$$\text{Res}(f; -3i) = \overline{\text{Res}(f; 3i)} = \frac{5}{4} - \frac{49}{12}i.$$

(a) The circle $|z-2|=2$ contains only the pole $z=1$. Therefore

$$\boxed{\int_C f(z)\,\mathrm{d}z = 2\pi i \cdot \frac{1}{2} = \pi i.}$$

(b) The circle $|z|=4$ contains all three poles. The sum of their residues is $\frac{1}{2}+\frac{5}{4}+\frac{5}{4}=3$. Hence

$$\boxed{\int_C f(z)\,\mathrm{d}z = 2\pi i \cdot 3 = 6\pi i.}$$

□

Exercise 81.5. Find the value of the integral

$$\int_C \frac{\mathrm{d}z}{z^3(z+4)},$$

taken counterclockwise around the circle (a) $|z|=2$; (b) $|z+2|=3$.

Solution. The integrand has a pole of order 3 at 0 and a simple pole at -4. Write $f(z)=\phi(z)/z^3$, where $\phi(z)=1/(z+4)$ is analytic and nonzero at 0. By the pole-residue formula in Sec. 80,

$$\operatorname{Res}(f;0) = \frac{\phi''(0)}{2!} = \frac{1}{2!}\,\frac{2}{(z+4)^3}\bigg|_{z=0} = \frac{1}{64}.$$

Also,

$$\operatorname{Res}(f;-4) = \frac{1}{z^3}\bigg|_{z=-4} = -\frac{1}{64}.$$

(a) The circle $|z|=2$ contains only 0. Hence

$$\boxed{\int_C \frac{\mathrm{d}z}{z^3(z+4)} = 2\pi i \cdot \frac{1}{64} = \frac{\pi i}{32}.}$$

(b) The circle $|z+2|=3$ contains both 0 and -4. Their residues cancel, so

$$\boxed{\int_C \frac{\mathrm{d}z}{z^3(z+4)} = 0.}$$

□

Exercise 81.6. Evaluate the integral

$$\int_C \frac{\cosh(\pi z)}{z(z^2+1)}\,\mathrm{d}z$$

when C is the circle $|z|=2$, described in the positive sense.

Solution. The integrand has simple poles at 0, i, and $-i$, all inside C. At the origin,

$$\operatorname{Res}(f;0) = \frac{\cosh(\pi z)}{z^2+1}\bigg|_{z=0} = 1.$$

Since $\cosh(\pi i)=\cosh(-\pi i)=\cos\pi=-1$, the residues at $z=\pm i$ are

$$\operatorname{Res}(f;\pm i) = \frac{\cosh(\pi z)}{2z^2}\bigg|_{z=\pm i} = \frac{-1}{-2} = \frac{1}{2}.$$

The residue sum is $1+\frac{1}{2}+\frac{1}{2}=2$. By Cauchy's residue theorem,

$$\boxed{\int_C \frac{\cosh(\pi z)}{z(z^2+1)}\,\mathrm{d}z = 2\pi i \cdot 2 = 4\pi i.}$$

□

Exercise 81.7. Use the theorem in Sec. 77, involving a single residue, to evaluate the integral of $f(z)$ around the positively oriented circle $|z| = 3$ when

(a) $f(z) = \dfrac{(3z+2)^2}{z(z-1)(2z+5)}$;

(b) $f(z) = \dfrac{z^3 e^{1/z}}{1+z^3}$.

Solution. All finite singularities of each function lie in $|z| < 3$. Thus, by the theorem in Sec. 77, with $w = 1/z$,

$$\int_{|z|=3} f(z)\ \mathrm{d}z = 2\pi i \operatorname{Res}\left(\frac{1}{w^2} f\left(\frac{1}{w}\right); 0\right).$$

(a) Direct simplification gives

$$\begin{aligned}\frac{1}{w^2} f\left(\frac{1}{w}\right) &= \frac{(3+2w)^2}{w(1-w)(2+5w)} \\ &= \frac{1}{w}\,\frac{(3+2w)^2}{(1-w)(2+5w)}.\end{aligned}$$

The second factor is analytic at $w = 0$ and has value $9/2$ there. Therefore

$$\boxed{\int_{|z|=3} f(z)\ \mathrm{d}z = 2\pi i \cdot \frac{9}{2} = 9\pi i.}$$

(b) Here

$$\frac{1}{w^2} f\left(\frac{1}{w}\right) = \frac{e^w}{w^2(1+w^3)} = \frac{h(w)}{w^2}, \qquad h(w) = \frac{e^w}{1+w^3}.$$

Since h is analytic at 0, the residue of $h(w)/w^2$ is $h'(0)$. Differentiation yields

$$h'(w) = \frac{e^w}{1+w^3} - \frac{3w^2 e^w}{(1+w^3)^2}, \qquad h'(0) = 1.$$

Hence

$$\boxed{\int_{|z|=3} f(z)\ \mathrm{d}z = 2\pi i.}$$

□

Exercise 83.5. Let C denote the positively oriented circle $|z| = 2$ and evaluate the integral

(a) $\displaystyle\int_C \tan z\ \mathrm{d}z$;

(b) $\displaystyle\int_C \frac{\mathrm{d}z}{\sinh(2z)}$.

Solution. We use Theorem 2 in Sec. 83: if $q(a) = 0$, $q'(a) \neq 0$, and $p(a) \neq 0$, then p/q has a simple pole at a with residue $p(a)/q'(a)$.

(a) For $\tan z = \sin z / \cos z$, the poles inside $|z| = 2$ are $z = \pm\pi/2$. Since $(\cos z)' = -\sin z$ and $\sin(\pm\pi/2) \neq 0$,

$$\operatorname{Res}(\tan z; \pm\pi/2) = \left.\frac{\sin z}{-\sin z}\right|_{z=\pm\pi/2} = -1.$$

Thus

$$\boxed{\int_C \tan z\ \mathrm{d}z = 2\pi i(-1-1) = -4\pi i.}$$

(b) The zeros of $\sinh(2z)$ are $z = n\pi i/2$ ($n \in \mathbb{Z}$). Only 0 and $\pm\pi i/2$ lie inside C. Each is simple because

$$\frac{\mathrm{d}}{\mathrm{d}z}\sinh(2z)\Big|_{z=n\pi i/2} = 2\cos(n\pi) = 2(-1)^n \neq 0.$$

Consequently,

$$\operatorname{Res}\left(\frac{1}{\sinh(2z)}; \frac{n\pi i}{2}\right) = \frac{(-1)^n}{2}.$$

The three residues sum to $\frac{1}{2} - \frac{1}{2} - \frac{1}{2} = -\frac{1}{2}$, and hence

$$\boxed{\int_C \frac{\mathrm{d}z}{\sinh(2z)} = -\pi i.}$$

□

Exercise 83.7. Show that

$$\int_C \frac{\mathrm{d}z}{(z^2-1)^2+3} = \frac{\pi}{2\sqrt{2}},$$

where C is the positively oriented boundary of the rectangle whose sides lie along the lines $x = \pm 2$, $y = 0$, and $y = 1$.

Suggestion. By observing that the four zeros of the polynomial $q(z) = (z^2-1)^2 + 3$ are the square roots of the numbers $1 \pm \sqrt{3}\,i$, show that the reciprocal $1/q(z)$ is analytic inside and on C except at the points

$$z_0 = \frac{\sqrt{3}+i}{\sqrt{2}} \qquad \text{and} \qquad -\overline{z_0} = \frac{-\sqrt{3}+i}{\sqrt{2}}.$$

Then apply Theorem 2 in Sec. 83.

Solution. Put $q(z) = (z^2-1)^2 + 3$. Its zeros satisfy

$$z^2 = 1 \pm \sqrt{3}\,i = 2e^{\pm i\pi/3}.$$

Taking square roots gives the four zeros

$$\pm\frac{\sqrt{3}+i}{\sqrt{2}}, \qquad \pm\frac{\sqrt{3}-i}{\sqrt{2}}.$$

Exactly the following two lie inside the given rectangle:

$$z_+ = \frac{\sqrt{3}+i}{\sqrt{2}}, \qquad z_- = \frac{-\sqrt{3}+i}{\sqrt{2}} = -\overline{z_+},$$

since their real parts lie between -2 and 2 and their imaginary parts equal $1/\sqrt{2} \in (0,1)$. The other two zeros lie below the real axis.

Since $q'(z) = 4z(z^2-1) \neq 0$ at each zero, these are simple poles of $1/q(z)$. By Theorem 2 in Sec. 83, using $z_+^2 - 1 = i\sqrt{3}$ and $z_-^2 - 1 = -i\sqrt{3}$,

$$\operatorname{Res}\left(\frac{1}{q}; z_+\right) = \frac{1}{4i\sqrt{3}\,z_+},$$
$$\operatorname{Res}\left(\frac{1}{q}; z_-\right) = -\frac{1}{4i\sqrt{3}\,z_-}.$$

Their sum is

$$\begin{aligned}\frac{1}{4i\sqrt{3}}\left(\frac{1}{z_+}-\frac{1}{z_-}\right) &= \frac{1}{4i\sqrt{3}}\left(\frac{1}{z_+}+\frac{1}{\overline{z_+}}\right)\\ &= \frac{1}{4i\sqrt{3}}\frac{2\operatorname{Re} z_+}{|z_+|^2} = \frac{1}{4i\sqrt{2}},\end{aligned}$$

where $\operatorname{Re} z_+ = \sqrt{3}/\sqrt{2}$ and $|z_+|^2 = 2$. Cauchy's residue theorem therefore gives

$$\boxed{\int_C \frac{\mathrm{d}z}{(z^2-1)^2+3} = 2\pi i \cdot \frac{1}{4i\sqrt{2}} = \frac{\pi}{2\sqrt{2}}.}$$

□